

Hallucination Analysis

This note explains how ClinicGuard ReportGen defines, reduces, and audits unsupported claims in generated radiology-style reports.

The project is a prototype. The included numbers are sample-artifact checks unless they are regenerated from a documented evaluation run.

1. Why Hallucination Control Matters

In medical report generation, a hallucination is a clinically meaningful statement that is not supported by the available evidence. In this project, evidence can come from:

- the image classifier confidence score,
- a Grad-CAM image region,
- structured patient history supplied by the user,
- a prior report supplied by the user.

The system is designed to make unsupported claims visible rather than hiding them inside fluent prose.

2. Hallucination Types

Common failure modes include:

- Fabricated finding: reporting a pathology that is not supported by image evidence.
- Anatomical shift: assigning a finding to the wrong region or side.
- Negation inversion: turning "no pneumothorax" into "pneumothorax present."
- Severity distortion: overstating the size or severity of a finding.
- Source mismatch: using patient history or prior text as if it were current visual evidence.

3. Mitigation Strategy

ClinicGuard uses three controls:

1. Confidence-gated findings: low-confidence findings are omitted or hedged.
2. Template-based generation: report text is assembled from controlled clinical phrases.
3. Evidence logging: generated claims are linked back to a source reference and checked after generation.

This does not guarantee clinical correctness, but it does make unsupported output easier to detect and review.

4. Sample Evidence Check

The bundled `reports/GROUNDING_EVIDENCE_LOG.csv` contains 10 sample claim rows. One row is marked as hallucinated and uses an `UNGROUNDED` source reference. From this sample artifact:

Sample Measure	Value
Claim rows	10
Flagged hallucinated rows	1
Sample hallucination flag rate	10%
Rows with non-ungrounded source references	9
Sample grounded-reference rate	90%

These values are useful for demonstrating the audit format. They are not a clinical performance benchmark.

5. Recommended Evaluation Standard

A credible benchmark should save:

- sample IDs,

- source images or stable image references,
- ground-truth labels,
- model confidence scores,
- generated reports,
- evidence logs,
- hallucination flags,
- exact command and configuration used for the run.

Without those artifacts, precision, recall, grounding, and hallucination metrics should be described as examples rather than final performance claims.

6. Practical Review Checklist

When reviewing generated output:

- Check every positive finding has either a grounded visual source or an explicitly named non-visual source.
- Check absent findings are not contradicted by high-confidence positive predictions.
- Check hedged findings are not presented as certain in the impression.
- Check prior-report language is not copied as a current image finding.
- Check each evidence row has a useful ``source_reference``.